

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

Aim:

To write a Prolog program to find the **number of vowels in a given string**.

Objective:

To identify vowels (a, e, i, o, u) from a string and count their total number using recursion.

Algorithm:

1. ☐ Start the program.
2. ☐ Read a string as a list of characters.
3. ☐ Take the first character.
4. ☐ Check whether the character is a vowel.
5. ☐ If it is a vowel, increment the count.
6. ☐ If it is not a vowel, continue without incrementing.
7. ☐ Recursively process the remaining characters.
8. ☐ When the list becomes empty, return the total count.
9. ☐ Stop.

Prolog Program:

```
% Program to count vowels
```

```
vowel(a).
```

```
vowel(e).
```

```
vowel(i).
```

```
vowel(o).
```

```
vowel(u).
```

```
count_vowels([], 0).
```

```
count_vowels([H|T], Count) :-
```

```
vowel(H),
```

```
count_vowels(T, Count1),
```

Count is Count1 + 1.

count_vowels([H|T], Count) :-

\+ vowel(H),

count_vowels(T, Count)..

Output

Result:

Thus, the Monkey Banana Problem is successfully represented using Prolog.